

Leakage-aware repeated random-subspace XGBoost improves top-gene stability in TCGA-BRCA transcriptomic models

Abstract

Feature selection performed outside resampling can inflate the apparent performance of transcriptomic biomarker models. Even when leakage is avoided, a single train-test split can produce unstable gene rankings because many correlated genes compete for similar predictive information. We evaluated a fold-contained repeated random-subspace XGBoost workflow in TCGA-BRCA expression data. Genes were repeatedly partitioned into small batches of 100, 250, 500 or 1000 genes, ranked by shallow XGBoost models within each training fold, aggregated across repeated local competitions and exported as Top2000 candidate panels. Seven classification tasks were studied: tumour versus normal, PAM50 subtype, PAM50 subtype after excluding PAM50 genes, IHC subtype, ER status, HER2 status and TNBC status. Across tasks, the best small-batch Top2000 panels retained internal five-fold cross-validation performance relative to fold-matched full Top2000 panels. The mean Top100 Jaccard across folds increased from 0.24 for full-space rankings to 0.57 for the best small-batch or consensus rankings. Small-batch screening also recovered more fold-stable BRCA/PAM50 prior genes in several tasks, with the largest mean gain observed for TNBC status at batch size 100. Baseline comparisons with logistic regression and elastic net, together with label-compatible METABRIC transfer, showed that the task signal was not specific to one classifier. These results support repeated fold-contained screening as a stability-audit and candidate-panel prioritisation strategy rather than as evidence for clinically validated biomarkers.

Introduction

High-dimensional transcriptomic models are often interpreted through ranked gene lists. This practice is vulnerable to two distinct problems. First, any variance filter, feature ranking or gene selection step performed before resampling can leak information from held-out samples into model development. Second, leakage-free analyses can still yield unstable top-ranked genes when correlated features compete for similar predictive roles.

This study addresses the second problem under a leakage-controlled design. We asked whether repeated local feature competition, performed entirely within training folds, could make high-ranked gene lists more reproducible while preserving predictive performance. The goal was not to introduce XGBoost as a superior classifier, but to use tree-based gain rankings as a practical stress test for feature-selection stability in transcriptomic biomarker workflows.

Materials and Methods

TCGA-BRCA expression data and clinical annotations were obtained from UCSC Xena. The processed matrix contained 1218 samples and 20530 genes. Seven prediction tasks were analysed: tumour versus normal, PAM50 subtype, PAM50 subtype with PAM50 genes excluded, IHC subtype, ER status, HER2 status and TNBC status. Binary tasks were evaluated primarily by AUC, and multiclass tasks by macro F1.

All feature selection was nested inside five-fold cross-validation. For each task and training fold, genes were randomly assigned to batches of 100, 250, 500 or 1000 genes. Shallow XGBoost models were trained within these local feature spaces for 25 repeated ranking epochs. Gene gains were aggregated within each fold, and fold-specific Top2000 candidate panels were then used for second-stage model evaluation.

Fold-matched full Top2000 panels provided the main internal comparator. High-rank reproducibility was quantified by Top100 Jaccard overlap across the five folds. A prior-union set of BRCA driver and PAM50 genes was used only as an interpretation control. These genes were not used for training, tuning or forced selection.

To place the workflow in context, selected Top30 panels were also evaluated with XGBoost, logistic regression and elastic net. METABRIC primary-tumour data were used for external transfer where labels were compatible. Tumour-normal transfer was not evaluated because the available METABRIC matrix contains primary tumours only.

Results

Small-batch panels retained internal performance

The best small-batch Top2000 panel was within 0.005 primary-metric units of the full Top2000 panel in six of seven tasks. Five tasks showed a positive best-small-batch delta relative to the full Top2000 comparator. The largest positive delta was observed for HER2 status (+0.0054), followed by PAM50 subtype without PAM50 genes (+0.0040), ER status (+0.0017), TNBC status (+0.0011) and tumour-normal classification (+0.0003). PAM50 subtype and IHC subtype showed small negative deltas (-0.0004 and -0.0013, respectively).

Repeated local competition improved high-rank stability

Full-space Top100 gene lists were unstable across folds, with a mean Top100 Jaccard of 0.24. In contrast, the best small-batch or consensus Top100 lists reached a mean Jaccard of 0.57. The improvement was observed across all seven tasks. ER status showed the highest best-ranking stability (0.657), while IHC subtype remained the most difficult task (0.475). These results indicate that the apparent identity of "top genes" depends strongly on the ranking procedure, even when performance is largely preserved.

Prior-gene recovery supported the interpretability signal

Small-batch screening recovered additional fold-stable BRCA/PAM50 prior genes relative to full Top2000 panels. The prior-union recovery heatmap showed consistent gains for receptor-status and subtype-related tasks. The largest mean gain was 19.8 genes for TNBC status at batch size 100. PAM50 subtype without PAM50 genes, as expected, showed minimal prior-union recovery because the defining PAM50 genes had been removed from the feature space.

Baseline and external analyses constrained the claim

Top30 panels retained signal under XGBoost, logistic regression and elastic net, suggesting that the classification results were not an artefact of one model family. In METABRIC transfer, label-compatible primary-tumour tasks produced full or consensus primary metrics ranging from 0.756 to 0.993. Consensus panels were close to full Top2000 panels, but were not uniformly superior. These analyses therefore support robustness of the task signal, while limiting the claim to stability auditing and candidate prioritisation.

Discussion

This work shows that a leakage-aware repeated random-subspace strategy can make transcriptomic top-gene rankings more stable without materially reducing internal classification performance. The main contribution is methodological: feature selection is kept within resampling, split-dependent ranking instability is made visible, and repeated local competition reduces the dominance of full-space masking effects.

The findings should be interpreted conservatively. The recovered genes are candidates for prioritisation, not validated biomarkers. External validation is label-limited, and METABRIC does not provide a tumour-normal counterpart for the downloaded expression matrix. Prognosis analyses were treated as exploratory because endpoint mapping and cross-cohort calibration introduce additional uncertainty.

Overall, repeated fold-contained screening provides a practical audit layer for transcriptomic machine-learning studies. It can help distinguish predictive tasks that are merely accurate from gene rankings that are also reproducible enough to justify biological follow-up.

Figure Legends

Figure 1. Leakage-aware repeated random-subspace screening. Feature ranking, panel export and validation are performed within resampling. Genes enter repeated local XGBoost competitions before Top2000 panel construction and stability assessment.

Figure 2. Small-batch screening improves Top100 reproducibility while retaining internal performance. a, Full-space Top100 Jaccard compared with the best small-batch or consensus Top100 Jaccard across five folds. b, Best small-batch Top2000 primary-metric delta relative to the fold-matched full Top2000 comparator. c, Prior-union recovery heatmap showing additional fold-stable BRCA/PAM50 signal recovered by small-batch screening.

Figure 3. Baseline and external validation context. a, Repeated internal Top30 classification performance for XGBoost, logistic regression and elastic net. b, METABRIC external primary metrics for full Top2000 and small-batch consensus panels in label-compatible primary-tumour tasks.

Data Availability

The analysis uses public TCGA-BRCA and METABRIC-derived expression and annotation data. Source-data tables, figures and selected Top30 gene panels are provided with the companion poster package.

Limitations

This is a candidate-panel study. The results do not establish causal biomarkers, clinical utility or mechanistic relevance. Biological validation and more extensive external testing are required before translational claims can be made.